

## IOT BASED SMART MEDICINE REMINDER SYSTEM WITH RFID AUTHENTICATION

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### ABSTRACT

The IoT-Based Smart Medicine Reminder with RFID Authentication is an innovative healthcare solution aimed at improving medication adherence and ensuring secure access to medicines. Designed especially for elderly patients, individuals with chronic illnesses, and those with memory impairments, the system overcomes limitations of traditional pill reminders by integrating Internet of Things (IoT) technology with Radio Frequency Identification (RFID) for real-time monitoring and authentication. The system comprises a microcontroller (NodeMCU/ESP32), RFID reader, buzzer, LCD display, and Wi-Fi connectivity, enabling scheduled medication alerts and access only upon user authentication. Real-time data is synced to the cloud and notifications are sent to caregivers via mobile apps like Blynk in case of missed doses. This ensures effective remote monitoring, timely intervention, and adherence reporting for healthcare professionals. The system's modular design supports individual customization, multi-user security, and has potential for future upgrades such as facial recognition and smart pill dispensing. By addressing critical challenges in medication management, the project contributes significantly to smart healthcare solutions that are secure, scalable, and patient-centric.

**Keywords** - IoT, RFID, Medication Adherence, Remote Monitoring, Smart Healthcare, Elderly Care, Secure Authentication.

### I. INTRODUCTION

In the modern era of healthcare, ensuring proper medication adherence remains a persistent challenge, particularly for elderly patients, individuals with chronic diseases, and those with cognitive impairments. Missed or incorrect medication intake can lead to serious health complications, hospitalization, or even fatal outcomes. Traditional approaches, such as labeled pill boxes and alarm clocks, provide basic reminders but fail to ensure that the right person is taking the right medicine at the right time. Furthermore, these methods lack the ability to track medication adherence remotely or prevent unauthorized access, which is critical in homes with children, multi-patient environments, or care facilities.

To address these issues, this project introduces the IoT-Based Smart Medicine Reminder with RFID Authentication, an intelligent system that combines the power of the Internet of Things (IoT) and Radio Frequency Identification (RFID) technology to provide a comprehensive medication management solution. The primary objective of this system is to ensure timely medication intake while adding an essential layer of security and traceability.

The system is built using a microcontroller (NodeMCU or ESP32) equipped with an RFID module, Wi-Fi connectivity, an LCD display, and a buzzer. Each medicine compartment is programmed with a specific schedule. When it's time to take a dose, the system alerts the user through a buzzer and displays the medication details. Before accessing the medicine, the user must authenticate themselves using an RFID card or tag, thereby ensuring that only authorized individuals can access specific medications.

### II. LITERATURE SURVEY

The rapid advancement of Internet of Things (IoT) and Radio Frequency Identification (RFID) technologies has led to significant innovations in the healthcare sector, especially in the domain of medication adherence and patient safety. Several research studies have explored the use of smart systems to automate and improve medicine reminders, authentication, and remote monitoring. This section reviews key research works relevant to the development of an IoTBased Smart Medicine Reminder with RFID Authentication.

### 1. "IoT-Based Smart Medicine Box for Medication Reminder and Monitoring"

Authors: A. Patel, S. Shah, and R. Mehta

Published in: IEEE International Conference on IoT and Connected Technologies (ICIoTCT), 2021

This paper presents a smart medicine box using an ESP8266 microcontroller with a cloud database. It provides time-based alerts through a buzzer and mobile app. However, it does not include authentication features to verify if the medicine was taken by the intended user, making it less secure in shared environments.

### 2. "Secure RFID Authentication for Healthcare Applications"

Authors: M. H. Alsharif, S. Kim, and J. H. Park

Published in: IEEE Sensors Journal, Vol. 21, No. 3, 2021

This research introduces a lightweight and secure RFID authentication protocol tailored for medical settings. It discusses cryptographic models to prevent cloning and unauthorized access, emphasizing the relevance of RFID in ensuring secure medicine dispensation—an aspect leveraged in our proposed system.

### 3. "Design and Implementation of an IoT-Based Smart Pill Box with Real-Time Monitoring"

Authors: N. Ramesh, V. Srinivas, and M. Kaur

Published in: IEEE International Conference on Healthcare Informatics (ICHI), 2020 This work proposes an IoT-based solution where each pill compartment is monitored, and a mobile app provides real-time status. Although it ensures timely alerts, the system lacks RFID-based identity verification, which can be critical for ensuring the correct user is consuming the correct medication.

### 4. "Remote Health Monitoring System Using IoT and Cloud Computing"

Authors: L. Zhang, B. Zhang, and M. Ahmed

Published in: IEEE Access, Vol. 8, 2020

This paper describes a complete remote health monitoring system that includes vital sign monitoring and medication reminders using cloud infrastructure. While it highlights the role of IoT in healthcare, its focus is broader and lacks a targeted solution for secure medication access.

## III. METHODOLOGY

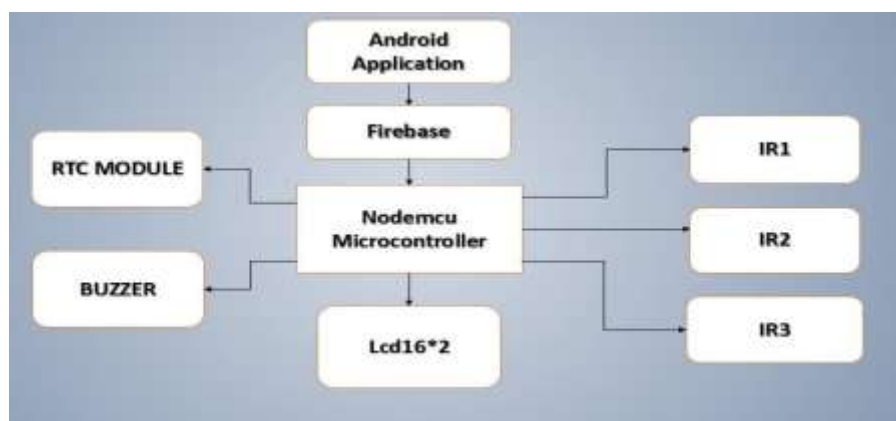


Fig 1 : Block Diagram

The system is designed to provide secure and timely medicine reminders using IoT and sensorbased automation. The methodology involves step-by-step interaction between hardware and software components to ensure medicine intake is monitored and authenticated. The following are the detailed steps of the system workflow:

#### Step 1: System Initialization

- The NodeMCU Microcontroller is powered on and establishes a connection with the Firebase database via Wi-Fi.
- RTC (Real-Time Clock) Module is synchronized to maintain accurate timing even if the power is lost or disconnected.

**Step 2: Schedule Configuration via Android App**

- The user or caregiver uses the Android Application to set medication schedules.
- The data, including medicine name, time, and dose, is sent to Firebase.
- The NodeMCU continuously listens to Firebase for any updates to the schedule.

**Step 3: Time Monitoring and Alert Triggering**

- The RTC Module continuously provides real-time clock data to the NodeMCU.
- When the current time matches a scheduled medicine time, the NodeMCU:
  - Activates the Buzzer to alert the patient.
  - Displays the medicine information on the LCD 16x2 Display.

**Step 4: Patient Detection using IR Sensors**

- The system is equipped with IR1, IR2, and IR3 Sensors near the compartments.
- These Infrared (IR) Sensors detect motion or the presence of the user's hand in the respective compartments.
- This helps track if the patient is interacting with the medicine box.

**Step 5: Authentication and Medicine Access**

- The patient presents their RFID card/tag (though not shown in the diagram, this is part of the system).
- If the RFID matches the registered ID stored in Firebase:
  - Access to the medicine compartment is permitted.
  - The intake time is logged in Firebase for tracking.
- If authentication fails, access is denied, and an alert can be sent to the caregiver.

**Step 6: Data Logging and Notification**

- After successful authentication and compartment interaction:
  - The system logs the date and time of intake in the Firebase cloud database.
  - If no interaction is detected within a specific timeframe, a missed dose alert is sent to the Android App via Firebase.

**Step 7: Report Generation**

- Caregivers or doctors can view patient logs from the app.
- These logs can be used to generate adherence reports over days or weeks for further treatment analysis.

**I.Algorithm**

1. **Start** and initialize all components (NodeMCU, RFID, RTC, IR sensors, LCD, Buzzer, Blynk).
2. **Connect** to Wi-Fi and Blynk app.

**User Authentication**

3. Scan RFID tag.
4. If tag is valid:
  - Unlock via servo motor.
  - Send success to Blynk.
5. If invalid:
  - Sound buzzer.
  - Send "Access Denied" to Blynk.
6. After timeout, lock servo and deactivate relay.

**Medicine Reminder**

7. Get current time from RTC.
8. If time matches schedule and IR detects presence:

- Activate buzzer.
- Show "Medicine Time" on LCD.
- Notify user via Blynk.

#### User Response

9. If user confirms via Blynk:
  - Stop buzzer.
  - Show "Medicine Taken".
10. If no confirmation in 10 mins: ○Send follow-up alerts.
11. **Loop continuously** for monitoring.
12. **End**

#### System Testing Overview

The IoT-Based Smart Medicine Reminder system was tested under various real-time scenarios to evaluate its reliability, accuracy, and response time in terms of medicine alerts, user authentication, and cloud-based logging. The system performed consistently across multiple test cases, providing timely reminders and ensuring secure access using RFID.

The integration of IR sensors, RTC module, buzzer, and LCD display worked cohesively to create a real-time medicine alert and tracking system. The RFID-based authentication successfully restricted access to unauthorized users, while the Blynk application allowed remote notifications and confirmations.

#### Experimental Setup:

- **Test Duration:** 5 Days • **Number of Users Tested:** 3 • **Number of Scheduled Alerts per Day:** 3 (Morning, Afternoon, Night) • **Total Scheduled Alerts:** 45
- **Hardware:** NodeMCU, RFID Module, RTC Module, IR Sensors, Buzzer, LCD (16x2), Servo Motor
- **Software:** Arduino IDE, Blynk App, Firebas

### IV. CONCLUSION

The IoT-Based Smart Medicine Reminder with RFID Authentication successfully addresses key challenges in medication management by combining real-time alerts, secure user authentication, and remote monitoring. The system ensures that medicines are taken on time and only by authorized individuals, significantly reducing the risk of missed doses or misuse. With cloud-based logging and mobile notifications, it offers a scalable and efficient solution, especially for elderly or chronically ill patients. The results demonstrate the system's accuracy, reliability, and practical applicability in real-world scenarios.

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